

BOUNDED GATE DEFECTS, FACTOR-SUM RETURNS AND FIXED-TAIL PELL FIBRES FOR ERDŐS–STRAUS TRACE SOURCES

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ABSTRACT. An original rational Erdős–Straus trace has an exact additive defect in a square-zero ideal modulo its residual. We retain the full bounded prime exponents and both channel types, derive their mixed composition law, and partition the complete original source into finite cyclic coefficient polynomials. A sharp unit-direction saturation theorem forces a full gate on specified original parts. Prime-square middle traces are terminal for both inherited residual-divisor descent systems, but a different factor-sum arrow has a complete quadratic inverse and a Pell specialization with verified hard-prime examples. An exhaustive certificate identifies 67369 as the least prime $1 \bmod 24$ at which the first trace precedes every full original gate; thus arbitrary-source repair cannot universally decrease the distinguished denominator. A second exact Pell lattice classifies every target that retains the numerical tail sum. A universal four-case argument proves that no proper prime-square trace has such a return at a proper divisor of its source residual; the complete certificate also proves that all 51 proper trace sums at $p = 67369$ have empty integral fibres. These are arithmetic source results, not post-witness spectral bounds or a universal existence theorem.

1. ARITHMETIC OBJECTIVE AND SOURCE BOUNDARY

The current programme does not yet force a positive integer witness at every prescribed prime. Its exact receiver and mixed-source fibre theorems begin with an occupied source. We return to the complete original bounded divisor boxes, rather than strengthening another post-witness norm estimate. The basis read for this continuation is the integrated workbench at `20709bee872ad10e4cc7d7b976d975e24532a803`. In particular, the redundant orientation bit removed during integration is not reinstated: an M word already records its tail order; an E word has its separately retained swap. The current corrections and the exact scope of the prior inverse are recorded in the integration audit (The Clankers, 2026a). The rational trace relaxation and the earlier residual-divisor maps are inherited from the pinned mod-24 source (The Clankers, 2026b, §§2–3).

The new results have three components. First, the full square-gate defect is an additive cyclic coordinate in an explicitly square-zero ideal. Its bounded multiplicative source has an exact decomposition into finite additive polynomials, with integer coefficients, stabilizers, and a sharp saturation criterion. Second, prime-power and prime-square sources admit exact classifications, including a terminal prime-square middle obstruction to all existing same-word and same-ray divisor deletions. Third, a different factor-sum map crosses some of those obstructions and has a complete inverse; on square carriers its precise arithmetic is a Pell equation. A finite least-example certificate proves that no universal repair from every trace can be required to decrease the original distinguished denominator.

No assertion of universal ES, uniform source occupancy, or historical priority is made. The term “square-zero” below refers to the specified finite-ring ideal. It is not an identification with an independently defined Split-Zero cohomology class or its support types. The connecting maps to the original integer coefficients are displayed instead.

2. ORIGINAL E/M BOXES AND RATIONAL TRACE COORDINATES

Fix a prime $p \equiv 1 \pmod{4}$ and an integer a with $p/4 < a < p/2$. Set $R = 4a - p$. Then $3 \leq R < p$, $R \equiv 3 \pmod{4}$, and $\gcd(pa, R) = 1$. Every original word is

$$a = \prod_{i=1}^n q_i^{e_i}, \quad u = \prod_i q_i^{f_i}, \quad 0 \leq f_i \leq 2e_i.$$

Write $\beta_i = f_i - e_i$ and normalize

$$g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g).$$

Thus $a = hrs$, $u = hr^2$, $\gcd(r, s) = 1$, and the centred prime coordinate is $\beta_i = v_{q_i}(r) - v_{q_i}(s)$. All following rational triples retain their order:

$$E : (a, y, z) = (hrs, hsk\kappa, phr\kappa), \quad \kappa = (pr + s)/R, \quad (1)$$

$$M : (a, y, z) = (hrs, phs\lambda, phr\lambda), \quad \lambda = (r + s)/R. \quad (2)$$

Their reciprocal sum is $4/p$, by multiplying through by $phrs\kappa$ or $phrs\lambda$. Equivalently the two tails are

$$E : \left(\frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \quad M : \left(\frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right).$$

The full gates are $R \mid 4u+1$ and $R \mid p+4u$, respectively. The ordered inverses are $u = a^2/(Ry - pa)$ and $u = pa^2/(Ry - pa)$. These identities are inherited, not newly discovered here: Yamamoto's Lemma 2, equation (4), printed p. 38 supplies the exterior congruences (Yamamoto, 1965); the general Type-I/II representations and their converses are in Elsholtz and Tao (2013, §2, Propositions 2.2 and 2.6).

An explicit crosswalk to their six coordinates is also available. Our full M state maps to their Type-II sextuple

$$(a_{ET}, b_{ET}, c_{ET}, d_{ET}, e_{ET}, f_{ET}) = (s, r, \lambda, h, R, 4hs\lambda - 1).$$

Their equation (2.22) returns our (a, y, z) without a permutation. Our full E state maps to their Type-I sextuple

$$(r, \kappa, s, h, (r + \kappa)/s, R),$$

and their projection returns (z, a, y) ; this cyclic permutation is retained. For rational traces these parameters need not all be integers.

Lemma 2.1 (Complete rational trace source). *Put $G_E = 4u+1$, $G_M = p+4u$. The corresponding two tails have integral sum exactly when $R \mid G_c^2$. Their common reduced denominator is $d = R/\gcd(R, G_c)$; on a trace, $d^2 \mid R$. Every positive rational pair (y, z) at this integral a satisfying*

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{y} + \frac{1}{z}$$

and having integral sum $y + z$ occurs in these coordinates, with the E swap retained.

Proof. The sums are $(a + pu)^2/(Ru)$ and $p(a + u)^2/(Ru)$. Since $u \mid a^2$ and all p, a, u are units modulo R , their integrality gives the square gates. The gcds of the two integral numerators with R agree and give the stated common denominator. Conversely, if $N = y + z \in \mathbb{Z}$, the reduced denominators of y, z coincide at d . Their product $yz = Npa/R$ has reduced denominator d^2 , since the reduced numerators of both tails are units at every prime dividing d . Hence $d^2 = R/\gcd(R, N)$, and $d \mid R$. The numbers $b = Ry - pa$, $c = Rz - pa$ are positive integers satisfying $bc = (pa)^2$. Their p -valuation pair is $(0, 2)$, $(1, 1)$, or $(2, 0)$. For $(0, 2)$ take $u = a^2/b$; for $(2, 0)$ first exchange the tails; for $(1, 1)$ take $u = pa^2/b$. These are precisely the original words and formulas. $\square$

Lemma 2.2 (Character parity). *For an original trace,*

$$E : (u/p) = -1, \quad M : (u/p) = -(a/p).$$

Here the symbols are Legendre symbols at p ; Jacobi symbols at R are used only inside the proof.

Proof. For odd $q_i \mid a$, reciprocity and $R \equiv -p \pmod{q_i}$ give $(q_i/R) = (p/q_i) = (q_i/p)$. At two the supplementary law and $R \equiv -p \pmod{8}$ give the same equality. Multiplication through all original exponents proves $(u/R) = (u/p)$ and $(a/R) = (a/p)$. The square gate implies the unsquared gate at every prime dividing R , not necessarily modulo R . It suffices for its Jacobi value, with all prime multiplicities retained. Since $(-1/R) = -1$ and $(p/R) = (R/p) = (a/p)$, the E normalized residue pu/a has Jacobi value (u/p) ; the M residue u/a has value $(u/p)(a/p)$. Both must equal -1 . $\square$

3. A SQUARE-ZERO ADDITIVE DEFECT ON THE ENTIRE SOURCE

Define

$$K = \prod_{\ell^v \parallel R} \ell^{\lceil v/2 \rceil}, \quad D = R/K, \quad \delta = R/D^2.$$

Then $R = KD = \delta D^2$, $D \mid K$, and δ is the positive squarefree part, not the radical. Also $\delta \equiv 3 \pmod{4}$, hence

$$D \leq \sqrt{R/3} < \sqrt{p/3}. \quad (3)$$

The ideal $J = K\mathbb{Z}/R\mathbb{Z}$ obeys $J^2 = 0$, because $R \mid K^2$. The map $\mathbb{Z}/D \rightarrow J$, $c \mapsto Kc$, is an additive isomorphism.

For channel exponent $\eta = 1$ in E and $\eta = 0$ in M, put

$$z_{\eta,\beta} = p^\eta \prod_i q_i^{\beta_i} = p^\eta u/a \quad \text{in } (\mathbb{Z}/R\mathbb{Z})^\times.$$

Trace is exactly $z \equiv -1 \pmod{K}$. Thus it has a unique coordinate

$$z = -1 + Kc, \quad c \in \mathbb{Z}/D. \quad (4)$$

The full gate is $c = 0$, and the actual common tail denominator is

$$d = \frac{D}{\gcd(D, c)}. \quad (5)$$

Indeed the two normalized residues are unit multiples of their original gates, so $\gcd(R, G_c) = \gcd(R, z + 1) = K \gcd(D, c)$. The case $D = 1$ is retained: its defect group is trivial and every trace is full.

Theorem 3.1 (Mixed integer-exponent law). *Take original traces $(\eta_j, \beta^{(j)}, c_j)$ at a common (p, a, R) . Let $n_j \in \mathbb{Z}$, with*

$$N = \sum_j n_j \text{ odd}, \quad \eta_* = \sum_j n_j \eta_j \in \{0, 1\}, \quad \beta_* = \sum_j n_j \beta^{(j)} \in \prod_i [-e_i, e_i].$$

Then this is an original trace in channel η_ , with*

$$u_* = a^{1-N} \prod_j u_j^{n_j}, \quad c_* = \sum_j n_j c_j \pmod{D}. \quad (6)$$

It is a full original state exactly when the latter sum is zero.

Proof. Write $z_j = -(1 - Kc_j)$. In the square-zero ideal, $(1 - Kc_j)^{n_j} = 1 - n_j Kc_j$ for every integer exponent, including negative ones. Their product, since N is odd, is $-1 + K \sum n_j c_j$. Its channel and prime exponents are precisely the displayed ones. The bounded exponent requirement proves that u_* is a positive integer divisor of a^2 , rather than an arbitrary unit residue. Apply (1)–(2) and their ordered inverses. $\square$

In particular $E_1 - E_2 + M$ returns M when its original box permits it:

$$u_* = u_{E,1} u_M / u_{E,2}, \quad c_* = c_{E,1} - c_{E,2} + c_M.$$

Retaining the two other inputs gives an inverse for this ternary map. For a general coefficient vector, the complete fibres are the bounded integer solutions of the displayed linear equations in the $\beta^{(j)}$ and their channel exponents, within their stated original trace domains; no assertion of injectivity after forgetting inputs is made.

3.1. Every coarse-neutral line has an exact finite return. Let $\gamma \in \mathbb{Z}^n$ satisfy $\phi(\gamma) = \prod q_i^{\gamma_i} \equiv 1 \pmod{K}$. Put

$$\lambda_\gamma = (\phi(\gamma) - 1)/K \pmod{D}.$$

The quotient is well-defined by first evaluating ϕ modulo R . For $\beta(t) = \beta_0 + t\gamma$ in the original box,

$$c(t) = c_0 - \lambda_\gamma t \pmod{D}. \quad (7)$$

The integer interval $L \leq t \leq U$ is the intersection of the inequalities $-e_i \leq \beta_{0i} + t\gamma_i \leq e_i$, using exact floors and ceilings. For $g = \gcd(\lambda_\gamma, D)$ there is no full state if $g \nmid c_0$. Otherwise, put $m = D/g$ and $t_0 = (c_0/g)(\lambda_\gamma/g)^{-1} \pmod{m}$ (take $t_0 = 0$ for $m = 1$). All full states are exactly $t = t_0 + jm$ in $[L, U]$, and their number is

$$\max\left(0, \left\lfloor \frac{U - t_0}{m} \right\rfloor - \left\lceil \frac{L - t_0}{m} \right\rceil + 1\right).$$

This proves termination on the given line. It does not assert that an arbitrary line contains a full period or even a trace.

3.2. A mixed return where every individual deletion fails. At $p = 48049$, take $a = 12019 = 7 \cdot 17 \cdot 101$, $R = 27$, $K = 9$, $D = 3$. The three proper traces are

channel	u	$\beta_{7,17,101}$	(h, r, s)	c
E	29189	$(-1, 1, 0)$	$(101, 17, 7)$	1
E	119	$(0, 0, -1)$	$(119, 1, 101)$	2
M	707	$(0, -1, 0)$	$(707, 1, 17)$	1.

All tails have common denominator three. $E_1 - E_2 + M$ lies in the original box and gives $u_* = 173417$, $\beta_* = (-1, 0, 1)$ and $c_* = 0$. Its marking is $(h, r, s, \lambda) = (17, 101, 7, 4)$, with

$$\frac{4}{48049} = \frac{1}{12019} + \frac{1}{22871324} + \frac{1}{330000532}. \quad (8)$$

For each individual trace, both the inherited fixed-word modulus $4K(u)$ and the ray modulus $4rs$ exceed R ; hence no proper residual-divisor deletion can be one modulo either modulus. The mixed operation changes the available word inside the old box instead. Its neutral-line version has $\gamma = (-1, 1, 1)$, $\lambda_\gamma = 1 \pmod{3}$, $t \in \{0, 1\}$, and $c(t) = 1 - t$. The actual short interval already contains its full state; no abstract subgroup replacement is needed.

4. COMPLETE BOUNDED ADDITIVE POLYNOMIALS AND SHARP SATURATION

For each original prime $q_i \mid a$, set $o_i = \text{ord}_K(q_i)$. Partition its actual interval $[-e_i, e_i] \cap \mathbb{Z}$ by residue modulo o_i . Every nonempty part has the unique form

$$\beta_i = b_i + o_i t_i, \quad 0 \leq t_i < N_i,$$

where b_i is its smallest member. Taking products of these parts partitions *every* original exponent vector once. Retain separately the channel.

A product part is a coarse trace part exactly when $p^\eta \prod q_i^{b_i} \equiv -1 \pmod{K}$. Let c_0 be its defect and put

$$\lambda_i = (q_i^{o_i} - 1)/K \pmod{D}.$$

Every point of that part has defect $c_0 - \sum_i \lambda_i t_i$. Consequently its number of full original words is

$$\text{coeff}_{c_0}(\mathcal{C}), \quad \mathcal{C} = \prod_i ([0] + [\lambda_i] + \cdots + [(N_i - 1)\lambda_i]) \in \mathbb{Z}[\mathbb{Z}/D]. \quad (9)$$

Here coeff_{c_0} extracts the coefficient of the basis element $[c_0]$. These are geometric factors with the original coefficient one per exponent, not binomial factors from artificially distinguished prime occurrences. The inverse reads β_i , its part b_i , and $t_i = (\beta_i - b_i)/o_i$; the original word is $\prod q_i^{e_i + \beta_i}$. Thus (9) is an exact arithmetic refinement of the full box.

The homomorphism on free abelian modules sends each original basis word to its defect-bin basis. Over its actual occupied bins, its kernel has the usual basis of differences of original words in the same fibre. An empty zero bin is different from a nonzero integer coefficient reducing to zero modulo a prime. No coefficient is reduced modulo D ; only its *index* is.

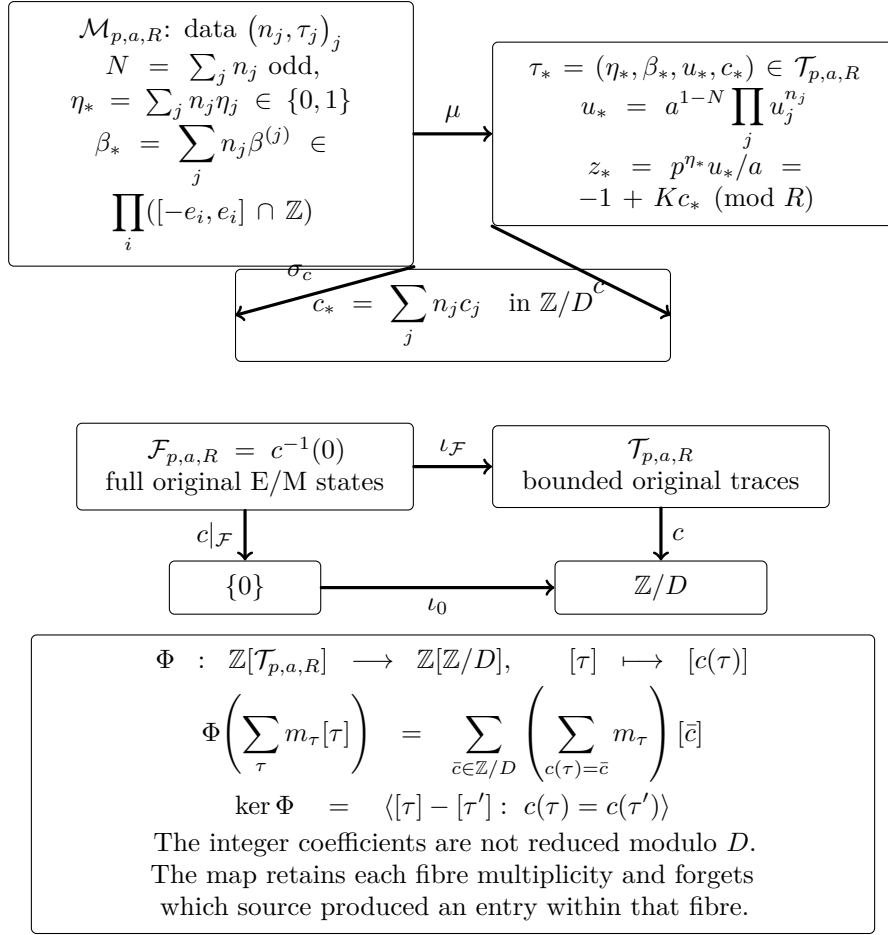


FIGURE 1. Typed maps for the defect equation $p^{\eta}u/a = -1 + Kc \pmod{R}$. In the upper triangle, $\sigma_c((n_j, \tau_j)_j) = \sum_j n_j c_j$, and the triangle commutes by the mixed integer-exponent law. The middle square identifies the full states as the zero fibre. The lower map states the exact kernel and the information lost when original trace words are replaced by their defect-bin coefficient vector.

Theorem 4.1 (Unit-direction saturation). *Suppose $D > 1$ and let $U_0 = \{i : \gcd(\lambda_i, D) = 1\}$. If*

$$\sum_{i \in U_0} (N_i - 1) \geq D - 1, \quad (10)$$

then every coefficient of $\mathcal{C}$ is at least $\prod_{i \notin U_0} N_i$. In particular the original box part contains at least that many full states. The number $D - 1$ cannot be lowered, even for packets inside actual hard-prime sources.

Proof. For a nonempty proper subset $S \subset \mathbb{Z}/D$ and a unit λ , $S + \lambda = S$ would make S invariant under a generator and hence the whole group. Therefore $|S \cup (S + \lambda)| \geq |S| + 1$. Adding successively $N_i - 1$ copies of each unit step expands the possible sumset until it is all of $\mathbb{Z}/D$ after at most $D - 1$ steps. This repeated-binary description is used only to prove equality of supports: the possible totals in coordinate i are still $0, \dots, N_i - 1$, each counted by the original geometric factor. Thus each residue has at least one original unit-coordinate vector. For each of the $\prod_{i \notin U_0} N_i$ choices of other coordinates, translate the required unit residue. The resulting full original vectors are distinct. A single interval of length $D - 1$ and unit step misses one residue, proving abstract sharpness. The next paragraph realizes that obstruction arithmetically. $\square$

Fix a prime $\ell \equiv 3 \pmod{4}$, put $R = \ell^3$, and choose a prime $q > 7$ with $q \equiv -1 + \ell^2 \pmod{\ell^3}$. This reduced progression contains infinitely many primes by Dirichlet's theorem (Sutherland,

2017, Theorem 18.1). Set $e = \ell - 1$. Choose primes $p > R$ satisfying

$$p \equiv 1 \pmod{840}, \quad p \equiv -R + 4q^e \pmod{q^{e+1}}.$$

These are compatible reduced conditions, since $q > 7$ and $q \nmid R$; another application of Dirichlet gives infinitely many such primes. An optional condition $p \equiv 1 \pmod{\ell}$ when $\ell \nmid 840$ remains reduced as well. Then $a = (p + R)/4 = q^e h$ with $q \nmid h$. Hold every other centred exponent at zero and vary $\beta_q \in [-e, e]$. Modulo $K = \ell^2$, the M traces are exactly the odd β_q . Modulo R , their residues are

$$q^{\beta_q} = -1 + \ell^2 \beta_q,$$

including negative exponents. Their defects run exactly through all nonzero classes modulo $D = \ell$. This original one-coordinate packet has $N = \ell - 1$, unit step $\lambda = -2$, and width $D - 2$, with no full word. Other words at the same shell may be occupied; none is deleted by this sharpness assertion.

4.1. Actual coefficient stabilizers. Write $C = \sum_j C_j[j]$. At frequency $k \in \mathbb{Z}/D$, its Fourier coefficient is nonzero exactly when, for every i ,

$$D \mid k\lambda_i \quad \text{or} \quad D \nmid k\lambda_i N_i.$$

Indeed a finite geometric sum is zero precisely when its ratio is nontrivial but its N_i th power is one. Let $\mathcal{F}$ be the displayed nonzero frequency set, and $g = \gcd(D, \{k : k \in \mathcal{F}\})$. Fourier inversion gives

$$\text{Stab}(C) = \{0, D/g, 2D/g, \dots, (g-1)D/g\}. \quad (11)$$

This is the stabilizer of the actual coefficient vector, not merely its support. It includes every multiplicity and is verified against direct shifts in the executable tables.

5. PRIME POWERS AND THE COMPLETE PRIME-SQUARE OBSTRUCTION

Proposition 5.1 (Prime-power M classification). *Let $p \equiv 1 \pmod{24}$, $a = q^e$, $q > 3$ prime, and suppose an M trace exists. Then $(q/p) = -1$. Write $\text{ord}_K(q) = 2b$. One has b odd and $q^b = -1 \pmod{K}$. Put*

$$d_b = R / \gcd(R, q^b + 1), \quad B = bd_b.$$

The complete centred trace exponents are the odd multiples of b in $[-e, e]$; the full exponents are the odd multiples of B . Their counts are

$$2 \left\lfloor \frac{e+b}{2b} \right\rfloor, \quad 2 \left\lfloor \frac{e+B}{2B} \right\rfloor.$$

At exponent bm with m odd, the common tail denominator is $d_b / \gcd(d_b, m)$.

Proof. Lemma 2.2 gives $(q/p)^{f-e} = -1$, so q is a nonresidue and $\beta = f - e$ is odd. In the cyclic subgroup generated by $q \pmod{K}$, the presence of -1 makes it the unique order-two element, reached at half the order. Hence every trace exponent is an odd multiple of b , and b is odd. The square-zero law gives $q^b = -1 + Kc$, with additive order of c equal to d_b . Thus $q^{2b} = 1 - 2Kc$ has order d_b , since D is odd, and $\text{ord}_R(q) = 2bd_b$. Also $q^{bd_b} = -1$ because d_b is odd. This proves the full exponent description and the denominator formula. Counting positive and negative odd multiples gives the two displayed counts. $\square$

Theorem 5.2 (Complete prime-square trace classification). *Let $p \equiv 1 \pmod{24}$, $q > 3$ prime and $a = q^2$ in the first-half range. A trace forces $q \equiv 2 \pmod{3}$. There are exactly four possible marked states:*

channel	u	trace condition	full condition
E	q	$K \mid 4q + 1$	$R \mid 4q + 1$
E	q^3	$K \mid 4q^3 + 1$	$R \mid 4q^3 + 1$
M	q	$K \mid q + 1$	$R \mid q + 1$
M	q^3	$K \mid q + 1$	$R \mid q + 1$

E and M traces coexist only at $R = 3$, where all four are full. Both E traces coexist only at $R = 3$ or 75 . At $R = 75$ this occurs exactly when $q \equiv 11 \pmod{15}$; the low E state is full only at $q \equiv 56$

(mod 75), and the high one only at $q \equiv 11 \pmod{75}$. Every proper M trace has both full original gates empty at its entire shell.

Proof. Since $q^2 \equiv 1 \pmod{24}$, $R \equiv 3 \pmod{24}$ and $3 \mid K$. If $q \equiv 1 \pmod{3}$ no word passes either coarse gate. Otherwise each gate forces an odd exponent f , leaving exactly $f = 1, 3$ in the full box $0 \leq f \leq 4$. For M these both reduce, by cancellation of a power of q , to $q + 1$, proving the table. If M and either E coexist, reduction at $q = -1 \pmod{K}$ forces $K \mid 3$; $R \equiv 3 \pmod{24}$ then gives $R = 3$. If both E states coexist, the identity

$$16(4q^3 + 1) - (4q + 1)(16q^2 - 4q + 1) = 15$$

forces $K \mid 15$. Enumerating the possible exponents of three and five (each at most two in R) with $R \equiv 3 \pmod{24}$ gives $R = 3, 75$. At 75 the five classes $q \equiv 11, 26, 41, 56, 71 \pmod{75}$ give the stated full-gate alternatives by substitution. A proper M trace cannot have $R = 3$, so no E trace coexists; its two M orientations are both nonintegral. $\square$

Theorem 5.3 (Terminality for the inherited deletion systems). *For a proper M trace of Theorem 5.2, neither original word has a same-word residual-divisor return. Neither corresponding primitive M ray has a ray-preserving residual-divisor return. These claims retain every divisor of R , not only its prime factors or squarefree part. For a proper low E trace $u = q$, the complete nontrivial same-word deletion set is $\{4q + 1\}$ if $4q + 1 \mid R$, and is empty otherwise.*

Proof. For $u = q^3$, the same-word modulus is $4q^2 > R$, so no proper divisor of R is one modulo that modulus. For $u = q$, both word and M -ray budgets are $4q$. Any permitted $k > 1$ divides $R \mid (q + 1)^2$ and is one modulo $4q$. Its complement $v = (q + 1)^2/k$ is an integer one modulo q and exceeds one: $v = 1$ would make k even. Write $k = 1 + qA$, $v = 1 + qB$ with $A \geq 4, B \geq 1$. Then $qAB + A + B = q + 2$, an impossibility. The same $4q$ budget controls the opposite primitive ray, so it is also excluded. For low E replace $q + 1$ by $m + 1$, $m = 4q$. If $k > 1$ divides $R \mid (m + 1)^2$ and $k \equiv 1 \pmod{m}$, then its complement is also one modulo m and exceeds one, because $R < 2q^2 < (m + 1)^2$. Writing the two factors as $1 + mA, 1 + mB$ forces $mAB + A + B = m + 2$, whence $A = B = 1$. Thus $k = 4q + 1$ is the sole possibility. If it divides R , write $R = (4q + 1)v$. Then $v \mid 4q + 1$ and the deletion has full target gate at v . Its distinguished denominator is $q(q - v)$ and its retained word is q , proving sufficiency. $\square$

6. A DIFFERENT ARITHMETIC ARROW: RETAIN THE FACTOR SUM

The terminality theorem does not prohibit changing both the numerical word and the primitive ray. The following arrow does exactly that. Its completeness is relative to the displayed residual δ , unchanged raw factor r , and opposite shifts of h and s ; it is not a classification of every possible source-changing map.

Theorem 6.1 (Complete factor-sum return and inverse). *Take any oriented original M trace*

$$a = hrs, \quad u = hr^2, \quad \gcd(r, s) = 1, \quad R = 4hrs - p = \delta t^2, \quad \delta t \mid r + s.$$

Its targets with residual δ , unchanged raw factor r , and

$$h' = h - \delta w, \quad s' = s + \delta w, \quad w \in \mathbb{Z}$$

are precisely the integer roots of

$$r(\delta w^2 + (s - h)w) = \frac{t^2 - 1}{4}. \quad (12)$$

For such a root set $c = \gcd(r, s')$. Every root gives positive integers h', s' and the full primitive middle marking

$$\begin{aligned} a' &= h'rs' = (p + \delta)/4, & u' &= h'r^2, \\ (h_M, r_M, s_M, \lambda_M) &= (h'c^2, r/c, s'/c, (r + s')/(c\delta)). \end{aligned} \quad (13)$$

Writing $\lambda' = (r + s')/\delta = c\lambda_M$, its ordered denominators are

$$(h'rs', ph's'\lambda', ph'r\lambda'). \quad (14)$$

For a proper trace, $a' < a$. There are no roots unless $r \mid (t^2 - 1)/4$. When that holds, there are at most two, tested by the square discriminant $(h - s)^2 + (R - \delta)/r$ and integer divisibility of $h - s \pm \sqrt{(h - s)^2 + (R - \delta)/r}$ by 2δ .

For any marked full primitive target $(p, \delta, h_0, r_0, s_0, \lambda_0)$ with squarefree residual δ , its complete incoming fibre is as follows. Choose each $c > 0$ with $c^2 \mid h_0$ and put

$$h' = h_0/c^2, \quad r = cr_0, \quad s' = cs_0.$$

For each integer

$$-h'/\delta < w < s'/\delta, \tag{15}$$

retain it precisely when

$$t^2 = 1 + 4rw(s' - h') - 4r\delta w^2$$

is a positive odd square, $\delta t^2 < p$, and

$$\gcd(r, s' - \delta w) = 1, \quad \delta t \mid r + s' - \delta w.$$

Recover $h = h' + \delta w$, $s = s' - \delta w$, $a = hrs$, $R = \delta t^2$, $u = hr^2$. Retaining (c, w) makes the two constructions inverse.

Proof. Multiplication gives $h'rs' = hrs + r\delta w(h - s) - r\delta^2 w^2$. Requiring $4h'rs' - p = \delta$ is precisely (12). Its target product $h's' = (p + \delta)/(4r)$ is positive and the factor sum remains $h + s > 0$; hence both factors are positive. Since $\delta \mid r + s$ and $s' = s + \delta w$, the full raw M gate holds. The old r is a unit modulo δ , so $c = \gcd(r, s')$ is also a unit there. Since $c \mid r + s'$, the quotient $(r + s')/(c\delta)$ is an integer. The displayed primitive normalization follows directly from c and gives exactly the same ordered denominators. The word $u' = h'r^2$ divides a'^2 , with complementary divisor $h's'^2$. Its original first-half range follows from $3 \leq \delta < p$. Properness implies $t > 1$ and $a - a' = \delta(t^2 - 1)/4 > 0$. The divisibility by r and discriminant test are exactly the quadratic formula, retaining every denominator.

Conversely, a primitive target has $h_0 = h'c^2$, $r_0 = r/c$, $s_0 = s'/c$; thus every lost gcd is among the finite list $c^2 \mid h_0$. For each, positivity of the source factors is (15). Multiplication gives exactly the displayed t^2 . The remaining finite tests are precisely source coprimality, the original trace gate, and the first-half range. They are therefore sufficient and necessary. Because $\gcd(r_0, s_0) = 1$, the raw target gcd is exactly c , so both compositions recover (c, w) and every original coordinate. $\square$

The source is oriented. Complementing the input word need not preserve this particular arrow; it changes which raw factor is fixed. An initial complement followed by the theorem and a target complement is a separately marked composite. It is not counted as an extra unconstrained orientation bit on an already ordered source. The theorem changes the numerical word and generally the primitive ray, so it is neither inherited deletion system.

At the hard prime $p = 2003761 \equiv 361 \pmod{840}$, take

$$a = 501034 = 2 \cdot 479 \cdot 523, \quad R = 375 = 15 \cdot 5^2, \quad u = 1916, \quad (h, r, s) = (479, 2, 523).$$

Both full original gates at this shell are empty. The source has common denominator five. Equation (12) has $w = -3$, yielding raw target $(h', r, s') = (524, 2, 478)$ and gcd $c = 2$. Its primitive marking is

$$(h_M, r_M, s_M, \lambda_M) = (2096, 1, 239, 16),$$

with

$$\frac{4}{2003761} = \frac{1}{500944} + \frac{1}{16060352806144} + \frac{1}{67198128896}.$$

The whole incoming fibre of this marked target has three (w, c) pairs: $(0, 1)$, $(-3, 2)$ and $(55, 4)$. Forgetting the gcd would lose the selected source and would incorrectly describe the other two preimages.

6.1. The prime-square specialization is exactly Pell. If $r = 1$ and $h = s = q$, equation (12) becomes

$$t^2 - 4\delta w^2 = 1. \quad (16)$$

This gives a completely explicit integer family, not only a test at an unidentified source. Take $\delta \geq 3$ squarefree, $\delta \equiv 3 \pmod{4}$, positive integers t, w satisfying (16), and $k \geq 1$. Put

$$q = \delta tk - 1, \quad p = 4q^2 - \delta t^2, \quad a = q^2, \quad (17)$$

then

$$h' = q - \delta w, \quad s' = q + \delta w, \quad \lambda' = tk + w. \quad (18)$$

One has $\gcd(q, \delta t) = 1$, $p > \delta t^2 > 0$, $p \equiv 1 \pmod{4}$, and the full ordered identity (14). To check the strict inequality, $q \geq \delta t - 1$ and $2(\delta t - 1)^2 - \delta t^2 > 0$ for $\delta \geq 3, t \geq 1$. Also $t > 2\sqrt{\delta}w$ makes $q > \delta w$. The source trace denominator is exactly

$$d = t / \gcd(t, k),$$

by $R / \gcd(R, q + 1)$ with $R = \delta t^2$. No prime assumption is needed for this integer identity. When the independently tested p, q are prime and $p \equiv 1 \pmod{24}$, it crosses the terminal prime-square trace locus of Theorem 5.3.

For $\delta = 3$ an explicit infinite sequence is

$$(t_0, w_0) = (1, 0), \quad \begin{pmatrix} t_{n+1} \\ w_{n+1} \end{pmatrix} = \begin{pmatrix} 7 & 24 \\ 2 & 7 \end{pmatrix} \begin{pmatrix} t_n \\ w_n \end{pmatrix}.$$

Direct multiplication preserves $t^2 - 12w^2 = 1$, and t_n strictly grows for $n \geq 0$. The matrix has order twelve modulo 840, verified by its twelve explicit powers; its entries can equally be multiplied by hand. For $k = 4$ and $n \equiv 1 \pmod{12}$, (17) gives $p \equiv 529 \pmod{840}$. This proves infinitely many *integer* parameters in that hard residue with actual solutions. It does not assert infinitely many prime values of the quadratic expression or simultaneous primality of q and p .

Three isolated pairs have deterministic primality certificates by full trial division:

t	w	k	q	p	a'
7	2	4	83	27409	6853
97	28	4	1163	5382049	1345513
1351	390	114	462041	853922067121	213480516781.

At the first pair, the original shell is $a = 6889$, $R = 147$, $u = 83$. Both original full gates are empty and every old word/ray deletion fails. The new factors are 77, 89, with $\lambda' = 30$, giving

$$\frac{4}{27409} = \frac{1}{6853} + \frac{1}{5635016310} + \frac{1}{63314790}. \quad (19)$$

All fields and the complete inverse w -fibre are serialized for each example. For the largest example the incoming fibre has five indices $w = 0, 4, 390, 776, 780$, not just the selected prime-square source.

The general factor-sum map also works off the square carrier. At $p = 12577$, $a = 3196$, $R = 207$, $(h, r, s) = (47, 1, 68)$, one has $\delta = 23$, $t = 3$, $w = -1$. The new factors are 70, 45, giving a full original state at residual 23. The signed value of w is essential; no positivity of the parameter w is imposed in Theorem 6.1.

7. A SECOND, DISTINCT FIBRE: RETAIN THE NUMERICAL TAIL SUM

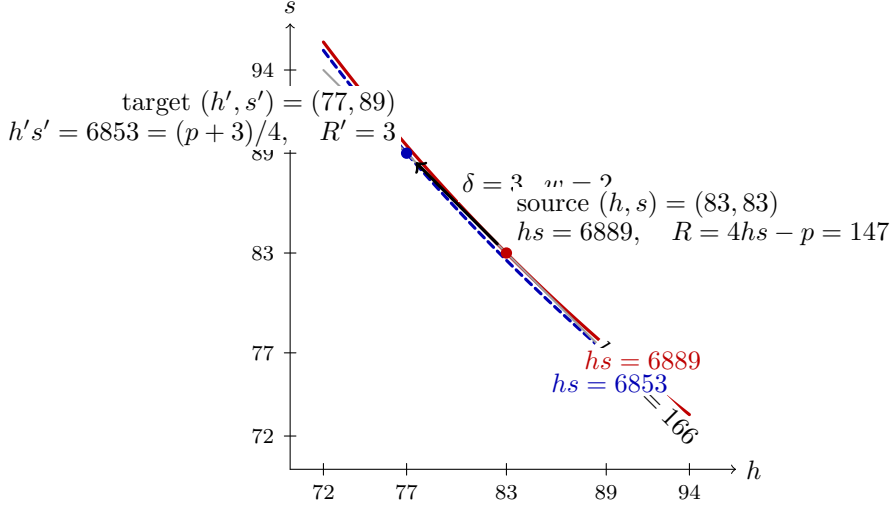
The preceding map retains the raw factor sum $h + s$. The next construction retains instead the numerical tail sum $Y + Z$. These are different functions on the original state space, so we keep separate symbols, domains and inverse conditions.

Theorem 7.1 (Complete fixed-tail-sum fibre). *Let p be an odd prime, let $N > p$, and put*

$$\mathcal{D} = N(N - p).$$

Ordered positive integer triples satisfying

$$\frac{4}{p} = \frac{1}{A} + \frac{1}{Y} + \frac{1}{Z}, \quad \frac{p}{4} < A < \frac{p}{2}, \quad Y + Z = N$$



$p = 27409$, $r = 1$, and $(h, s) \mapsto (h - \delta w, s + \delta w)$ sends $(83, 83)$ to $(77, 89)$ while retaining the factor sum $h + s = 166$. Both product levels and both residuals are displayed; the arrow does not identify the two hyperbolas.

FIGURE 2. The exact factor-sum arrow $(h, s) \mapsto (h - \delta w, s + \delta w)$ on the first Pell-family example. The source lies on $hs = 6889$ and the target lies on $hs = 6853$; their common line is $h + s = 166$.

are in bijection with pairs $(\rho, \Delta) \in \mathbb{Z}_{>0} \times \mathbb{Z}$ such that

$$\begin{aligned} \rho \mid N, \quad 0 < \rho < p, \quad \rho \equiv -p \pmod{4}, \\ \Delta^2 = \mathcal{D} - \frac{Np^2}{\rho}, \quad |\Delta| < N, \quad \Delta \equiv N \pmod{2}. \end{aligned} \quad (20)$$

The two mutually inverse coordinate maps are

$$\rho = 4A - p, \quad \Delta = Z - Y; \quad A = \frac{p + \rho}{4}, \quad Y = \frac{N - \Delta}{2}, \quad Z = \frac{N + \Delta}{2}. \quad (21)$$

Consequently this fibre is decided by a finite divisor-and-square test.

Write uniquely $\rho = \delta v^2$, with δ positive and squarefree, and put $X = v\Delta$. The same fibre is the union of the exact lattice sections

$$\begin{aligned} X^2 - \mathcal{D}v^2 = -\frac{Np^2}{\delta}, \quad \delta v^2 \mid N, \\ X \equiv Nv \pmod{2v}, \quad 0 < \delta v^2 < p, \quad \delta v^2 \equiv -p \pmod{4}. \end{aligned} \quad (22)$$

Every condition in (22) is part of the return map. An integer point on the displayed norm equation without the divisibility and congruence conditions need not give integer tails.

Proof. Set $\rho = 4A - p$. The reciprocal identity and $Y + Z = N$ give

$$YZ = \frac{NpA}{\rho} = \frac{Np(p + \rho)}{4\rho}.$$

Therefore

$$(Z - Y)^2 = N^2 - 4YZ = N(N - p) - \frac{Np^2}{\rho},$$

or, without a quotient,

$$\rho\Delta^2 = N((N - p)\rho - p^2). \quad (23)$$

The first-half inequalities give $0 < \rho < p$, while integrality of A gives $\rho \equiv -p \pmod{4}$. Since p is prime and $\rho < p$, one has $\gcd(\rho, p) = 1$; equation (23) then forces $\rho \mid N$. Positivity and integrality of Y, Z are exactly the final two conditions in (20). This proves necessity. Substitution

of (21) proves the reciprocal identity and every range condition, so it also proves sufficiency and both inverse compositions.

After $\rho = \delta v^2$, multiplication of the square equation by v^2 gives (22). Conversely $v \mid X$ and $X \equiv Nv \pmod{2v}$ recover an integer $\Delta = X/v$ with the required parity. The residual and range gates then recover exactly (21). Thus the Pell lattice description loses no coordinate and adds no candidate. $\square$

There is an equivalent shifted-factor conic which makes the missing gate visible. Put

$$B = \rho Y - pA, \quad C = \rho Z - pA, \quad w = B - C.$$

Then

$$BC = (pA)^2, \quad w^2 = \mathcal{D}\rho^2 - Np^2\rho. \quad (24)$$

With

$$U = 2\mathcal{D}\rho - p^2N, \quad V = 2w,$$

equation (24) becomes

$$U^2 - \mathcal{D}V^2 = p^4N^2. \quad (25)$$

However, exact integer tails require the additional gate

$$\rho \mid w, \quad Y = \frac{1}{2} \left(N + \frac{w}{\rho} \right), \quad Z = \frac{1}{2} \left(N - \frac{w}{\rho} \right). \quad (26)$$

Thus the unrestricted Pell orbit in (25) is a larger space of shifted-factor points. Its failure to land in (26) defines the exact return defect.

Theorem 7.2 (Prime-square fixed-sum divisor obstruction). *Let p, q be odd primes satisfying*

$$p \equiv 1 \pmod{4}, \quad 2q^2 < p < 4q^2, \quad R = 4q^2 - p.$$

For $u \in \{q, q^3\}$ put

$$G_E(u) = 4u + 1, \quad G_M(u) = p + 4u.$$

Choose $c \in \{E, M\}$ and suppose that the corresponding prime-square source is a proper trace:

$$R \mid G_c(u)^2, \quad R \nmid G_c(u).$$

Let (y, z) be its ordered rational tails from (1) or (2), and set $N = y + z \in \mathbb{Z}$. There is no positive integral target

$$\frac{4}{p} = \frac{1}{A} + \frac{1}{Y} + \frac{1}{Z}, \quad Y + Z = N,$$

whose residual $S = 4A - p$ is a proper divisor of the source residual:

$$S \mid R, \quad 0 < S < R.$$

The target may change the word, the E/M channel and the tail order.

Proof. The source data can be kept without suppressing any of the four marked states. Define H by the following table:

(c, u)	H	(y, z)	N
(E, q)	$p + q$	$(q^2H/R, pqH/R)$	qH^2/R
(E, q^3)	$pq + 1$	$(qH/R, pq^2H/R)$	qH^2/R
(M, q)	$q + 1$	$(pq^2H/R, pqH/R)$	pqH^2/R
(M, q^3)	$q + 1$	$(pqH/R, pq^2H/R)$	pqH^2/R

(27)

The trace gates transfer to H because

$$\begin{aligned} H &= qG_E(q) - R & (E, q), \\ H &= G_E(q^3) - qR & (E, q^3), \\ G_M(q) &= 4qH - R & (M, q), \\ G_M(q^3) &= qR + pH & (M, q^3). \end{aligned}$$

Every displayed multiplier is a unit modulo R , since $\gcd(R, pq) = 1$. Hence

$$R \mid H^2, \quad R \nmid H.$$

Put

$$g = \gcd(R, H), \quad \eta = R/g, \quad F = H/g.$$

Then $\gcd(\eta, F) = 1$. The divisibility $R \mid H^2$ forces $\eta \mid g$; write

$$g = k_0\eta, \quad R = k_0\eta^2, \quad H = k_0\eta F. \quad (28)$$

Here k_0 is a local gcd factor and is not the least square modulus K used in Sections 3–5. Properness is exactly $\eta > 1$. Since R is odd, $\eta \geq 3$, and

$$\gcd(\eta, pqF) = 1.$$

In every row of (27), neither p nor q divides H ; also $k_0 \mid R$. Hence

$$\gcd(pq, k_0F) = 1. \quad (29)$$

The four source sums in (27) therefore become

$$N = \begin{cases} qk_0F^2, & c = E, \\ pqk_0F^2, & c = M, \end{cases} \quad \text{and} \quad \gcd(R, N) = k_0. \quad (30)$$

Assume that a target with $S \mid R$ and $0 < S < R$ exists. Its first denominator satisfies the exact chain

$$A = \frac{p+S}{4} < \frac{p+R}{4} = q^2 < \frac{p}{2}.$$

In particular $p \nmid A$. Also $S \equiv -p \equiv 3 \pmod{4}$, so S is odd and $S \geq 3$. From the target identity and $Y + Z = N$,

$$YZ = \frac{NpA}{S}.$$

One has $\gcd(S, pA) = 1$: a common divisor of S and A divides $4A - S = p$, while $S < p$. Thus $S \mid N$, and

$$S \mid \gcd(R, N) = k_0. \quad (31)$$

The complete inverse in Lemma 2.1, applied at residual S , shows that after the required E tail orientation the target has exactly one of the following two forms, with $\gcd(r, s) = 1$:

$$E : \quad A = hrs, \quad S\kappa = pr + s, \quad (Y, Z) = (hsk, phr\kappa), \quad N = hS\kappa^2, \quad (32)$$

$$M : \quad A = hrs, \quad S\lambda = r + s, \quad (Y, Z) = (phs\lambda, phr\lambda), \quad N = phS\lambda^2. \quad (33)$$

All displayed parameters are positive integers. Because $p \nmid hS$, the p -adic valuation of an E target sum is even and that of an M target sum is odd. Equation (30) has valuation zero in the source E channel and one in the source M channel. Therefore the target channel equals the source channel.

Let $\theta = \kappa$ in E and $\theta = \lambda$ in M. Comparing target and source sums and using (31) gives

$$q \frac{k_0}{S} F^2 = h\theta^2. \quad (34)$$

The prime q divides neither (k_0/S) nor F , while $h \mid A < q^2$. Taking q -adic valuations in (34) yields

$$h = qh_0, \quad q \nmid h_0\theta.$$

Consequently $q \mid A$. There is a unique integer j with

$$A = q(q - j), \quad 0 < j < q, \quad R - S = 4qj, \quad (35)$$

and cancellation of q in (34) gives

$$\frac{k_0}{S} F^2 = h_0\theta^2. \quad (36)$$

There are three cases after the two M orientations are kept together.

Case $E, u = q$. Here $H = p + q = k_0\eta F$. From (36),

$$\kappa \leq F\sqrt{\frac{k_0}{S}}, \quad S\kappa \leq F\sqrt{Sk_0} \leq Fk_0 = \frac{p+q}{\eta} < p.$$

But the exact E gate in (32) gives $S\kappa = pr + s > p$, a contradiction.

Cases $M, u = q$ and $M, u = q^3$. In both cases $H = q + 1 = k_0\eta F$, hence $q = k_0\eta F - 1$. Using (35),

$$k_0\eta^2 - S = 4j(k_0\eta F - 1),$$

or

$$k_0\eta(\eta - 4jF) = S - 4j. \quad (37)$$

The right side is nonzero because S is odd. Moreover $0 < S \leq k_0 < k_0\eta$. Since k_0 and η are odd while $q + 1 = k_0\eta F$ is even, F is even and $k_0F \geq 2$, so $q \geq 2\eta - 1 > \eta$. It follows that

$$0 < 4j = \frac{R - S}{q} < \frac{R}{q} = \frac{k_0\eta^2}{q} < k_0\eta.$$

Thus $0 < |S - 4j| < k_0\eta$, contradicting (37), which makes $S - 4j$ a nonzero multiple of $k_0\eta$.

Case $E, u = q^3$. Here $H = pq + 1 = k_0\eta F$. Reduce κ/F to lowest terms by writing

$$F = nw, \quad \kappa = mw, \quad \gcd(m, n) = 1.$$

Equation (36) gives a positive integer τ such that

$$\frac{k_0}{S} = \tau m^2, \quad h_0 = \tau n^2.$$

Set $\Gamma = \tau m\eta n$. Then the source identity and the exact target gate give

$$pq + 1 = k_0\eta F = \Gamma S\kappa = \Gamma(pr + s),$$

so

$$\Gamma s \equiv 1 \pmod{p}. \quad (38)$$

From (35), $q - j = h_0rs = \tau n^2rs$, and hence

$$0 < \tau ns = \frac{q - j}{nr} < q.$$

Also

$$R = S\tau(m\eta)^2 < 2q^2.$$

The already retained congruence $S \equiv 3 \pmod{4}$ gives $S \geq 3$; therefore

$$(m\eta)^2 < \frac{2q^2}{3} < q^2.$$

It follows that

$$0 < \Gamma s = (m\eta)(\tau ns) < q^2 < p.$$

Together with (38), this forces $\Gamma s = 1$, whereas $\Gamma s \geq \eta \geq 3$. This final contradiction exhausts all four marked prime-square sources. $\square$

The divisor hypothesis in Theorem 7.2 is part of its scope. It does not exclude a smaller residual $S < R$ with $S \nmid R$, and it does not apply after changing the numerical tail sum N . It is also separate from the composite-carrier obstruction below.

At $p = 67369$, the proper M trace

$$(a, R, u, h, r, s) = (16849, 27, 1421, 29, 7, 83)$$

has ordered rational tails

$$(y, z) = \left(\frac{1621571830}{3}, \frac{136759070}{3} \right), \quad N = 586110300.$$

Its shifted factors are

$$(B, C, w) = (13459046189, 95731349, 13363314840), \quad w \equiv 18 \pmod{27}.$$

It therefore supplies an exact integer point of (25), but not an integer-tail return. The complete list allowed by (20) before the square test is

$$\rho \in \{3, 15, 75, 87, 435, 2175\}.$$

For $\rho = 3$ the discriminant is negative. For the remaining five residuals it is respectively 2 modulo 3, 5 modulo 7, 2 modulo 3, 3 modulo 7, and 2 modulo 3. Each is a quadratic nonresidue, so this source has no integral target anywhere in its fixed numerical tail-sum fibre.

The complete first-half census at this prime has 69 proper original trace tags with 51 distinct numerical tail sums, and 47 integral tags with 37 distinct tail sums. The two sets of sums are disjoint. Applying Theorem 7.1 to all 51 proper sums produces 507 eligible residual divisors; 478 have nonnegative discriminant and none has a square discriminant. The separately written checker recomputes the divisor lists, tail sums and square tests without importing the main implementation. This is a fixed-prime structural obstruction to retaining $Y + Z$, not a counterexample to Erdős–Straus and not a failure of the factor-sum map of Theorem 6.1.

8. A LEAST OBSTRUCTION TO UNIVERSAL NONINCREASING TRACE REPAIR

Theorem 8.1 (Finite least example). *Let $A_{\text{tr}}(p)$ and $A_{\text{full}}(p)$ be the least first-half distinguished denominators supporting, respectively, an original trace and an original full state; use $+\infty$ for an empty set. Among primes $p \equiv 1 \pmod{24}$, the least with $A_{\text{tr}}(p) < A_{\text{full}}(p)$ and $A_{\text{tr}}(p) < +\infty$ is*

$$p = 67369.$$

Its first trace shell has $a = 16849 = 7 \cdot 29 \cdot 83$, $R = 27$; its first full shell has $a = 16850$, $R = 31$. Thus no universal map from every original rational trace to an integral witness at the same prime can require $a' \leq a$.

Deterministic finite proof. The certificate enumerates all 817 primes congruent to one modulo 24 through 67369 by a sieve and independently verifies that list. For each it enumerates consecutive original shells from $R = 3$ through the first full shell, factoring a and listing the complete divisors of a^2 . For every smaller prime, the first trace shell already has a full gate; thus every trace at that prime has some full witness with distinguished denominator no larger than its own. Exactly at 67369 the first trace precedes the first full gate.

For that prime, every word and both gate residues are serialized for $R = 3, 7, 11, 15, 19, 23, 27, 31$. The first six have no traces. At $R = 27$, the four E trace words are 29, 83, 488621, 1398467, and the four M trace words are 1421, 4067, 69803, 199781. All are proper. At $R = 31$, $a = 16850 = 2 \cdot 5^2 \cdot 337$ has the two E words 674, 84250. The full enumeration and the independent rational-tail test agree on these lists.

The short first-half completeness argument below ensures that this excludes all integral witnesses with smaller distinguished denominator, not merely a preferred subfamily of states. Integer sieve, factorization and finite loops make the computation an executable exhaustive proof with the stated bound. No leastness claim beyond that checked prime class is made. $\square$

For completeness, every positive integer witness at an odd prime $p \equiv 1 \pmod{4}$ has a first-half distinguished denominator. At least one and at most two denominators are divisible by p . With two, the remaining unit denominator has reciprocal strictly greater than $2/p$, hence lies below $p/2$; it is above $p/4$ by positivity. With one, write that denominator pk and the other two $a = gr \leq b = gs$, $\gcd(r, s) = 1$. They are units at p . The equation gives $p \mid 4k - 1$, so $4k - 1 = pt$ with $t \equiv 3 \pmod{4}$. It reduces to $tgrs = k(r + s)$. Hence $k = jrs$, $r + s = t\lambda$, and $g = j\lambda$, since $\gcd(t, k) = 1$. If $a \geq p/2$, then $2jr(r - s) \geq -1$; its even nonpositive left side forces $r = s = 1$, contradicting $t \mid 2$. Thus $p/4 < a < p/2$. The factor-pair argument of Lemma 2.1, now with integral tails, then recovers a full original E/M word.

The least-example theorem does not contradict the successful strict descents above. It says that their domains cannot be extended to every trace merely by improving their choice of descending word. At this prime a general repair must sometimes move to a larger distinguished denominator, or start from a different trace already above the first occupied shell.

9. COUNTERFACTUAL CONSTRAINTS AND THE NEXT EXACT OBLIGATION

For an arbitrary prescribed prime $p \equiv 1 \pmod{24}$, the construction of Sections 3–4 is finite on every original shell. If all full gates failed, then every squarefree shell would have no trace; every coarse trace part at a squareful shell would have zero coefficient (9); in particular its unit-direction width would be at most $D - 2$. Every oriented M trace would fail the exact integer-root test (12), because a passing root constructs a genuine full state at the same prime. On a prime-power M source, $e < bd_b$ would be necessary. These are proved simultaneous necessary conditions in the original bounded arrays. They do not contradict one another by assumption.

The next missing implication is an arithmetic incompatibility of those complete zero coefficients across different shells, or another explicit source-changing mechanism that forces a full coefficient. The prefix example rules out universal nonincrease of a as its sole termination invariant. Receiver nonvanishing is not invoked as initial-source positivity. Neither a nonempty subgroup nor a limit of averages is substituted for an original word.

10. COMPUTATION, REPRODUCIBILITY AND SCOPE

The main standard-library verifier scans every original word at all 143 primes $p \equiv 1 \pmod{24}$ through 10000: 171732 shells and 5616298 words. Its exact trace counts are 5788 E and 5678 oriented M; full counts are 3637 E and 3910 M. It reconstructs 4139 additive parts over 2522 squareful trace shells and compares all coefficients with direct original gates. The prime-square scan through $p \leq 2000000$ contains 167 occupied trace shells and 285 marked states: E has 52 full and 83 proper; M has 56 full and 94 proper. These are source-state counts, not new primes solved globally. The prefix proof covers 817 primes and 1422 tested shells. The main prime-square lane generates candidate residuals from the three explicit squared linear forms; the separate lane instead scans the whole prime interval for each prime square and tests rational tail sums.

The separate implementation imports neither the main verifier nor its arithmetic module. It checks the full source census, every additive part, all prime-square sources, every prefix gate and every displayed arrow and its incoming fibre. It is a separately written check, not independent human peer review. Run `python run_all.py -directory reproduced` for normal, optimized, and separate lanes. Every failure raises an exception even under optimized Python; integer coefficients are never stored as floating-point numbers. The manifest and execution receipt record actual file hashes and observed runs. The universal proofs use the displayed ring, valuation, finite sumset and quadratic identities. Finite tests corroborate those proofs; the finite leastness statement is explicitly computational.

No Lean build, historical-priority determination or universal ES proof is claimed. The supplied Lean plan lists exact theorem dependencies. The original source, ordered maps, finite fibres and all exceptional tests remain available in `arithmetic.py`, `MORPHISMS.md`, and the JSON tables.

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